

STA 35C: Statistical Data Science III

Lecture 22: Principal Component Analysis

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Agenda

Quick recap: smoothing splines

- Why the solution is a natural cubic spline
- What effective degrees of freedom mean

Today:

- Unsupervised learning
 - Supervised vs. unsupervised learning
 - What unsupervised learning is for
- Principal component analysis (PCA): Intuition
 - PCA as dimension reduction
 - PCA in $p = 2$: projection and maximum variance
 - Why centering and standardization matter

Recap: Smoothing splines

Smoothing spline: estimate a smooth function g by solving

$$\min_{g \in \mathcal{G}} \left\{ \underbrace{\sum_{i=1}^n (y_i - g(x_i))^2}_{\text{RSS; data fidelity}} + \lambda \underbrace{\int (g''(t))^2 dt}_{\text{smoothness penalty}} \right\}$$

- The first term rewards fitting the observed data.
- The second term penalizes curvature, discouraging excessive wiggles.
- $\lambda \geq 0$ controls the tradeoff between fit and smoothness.
 - $\lambda = 0$: interpolation
 - $\lambda \rightarrow \infty$: least-squares line

Although the optimization ranges over many smooth functions g , the minimizer has a special form: it is a **natural cubic spline** with knots at the observed x_i 's.

- This does *not* mean we manually choose n knots and then fit an ordinary regression spline.
- The penalty term shrinks the fit toward smoother curves, so the *effective* flexibility can be far smaller than the number of knots suggests for (unpenalized) regression splines.

Recap: Effective degrees of freedom

Key point: A smoothing spline solution lives in a natural cubic spline space with knots at the observed x_i 's, but the penalty controls how flexible the fitted curve actually is.

Effective degrees of freedom for smoothing splines:

- A natural cubic spline with K (interior) knots has $K + 2$ degrees of freedom.
- If the knots are placed at all observed x_i 's, then there are $K = n - 2$ interior knots.
- Thus the full natural-spline space has $(n - 2) + 2 = n$ degrees of freedom (free parameters).
- Despite n nominal degrees of freedom, these are constrained/shrunk down

$$\begin{aligned}\lambda = 0 &\Rightarrow \text{df}_\lambda = n \quad (\text{interpolation}), \\ \lambda \rightarrow \infty &\Rightarrow \text{df}_\lambda = 2 \quad (\text{least-squares line}).\end{aligned}$$

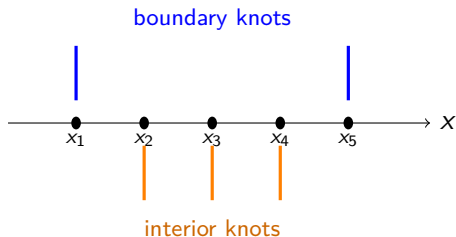
For $0 < \lambda < \infty$, df_λ lies between 2 and n .

For more technical details, see [JWHT21, Chapter 7.4.4 & 7.5.2].

Recap: Further elaboration on the df and the effective df

Assume $x_1, \dots, x_n$ are distinct and ordered.

Smoothing-spline fact: The solution is a natural cubic spline with boundary knots at x_1, x_n and interior knots at $x_2, \dots, x_{n-1}$.



For $n = 5$:

- Interior knots: $K = n - 2 = 3$.
- Natural cubic spline df:

$$K + 2 = (n - 2) + 2 = n.$$

- Thus the full natural-spline space has df n , not $n + 2$.

Effective df: The penalty controls how much of this n -nominal df is actually used.

Illustration of smoothing splines

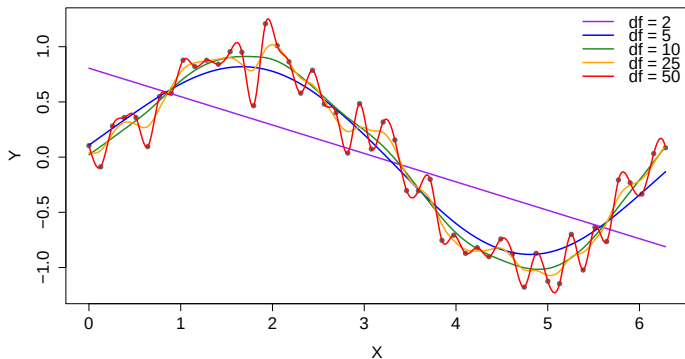


Figure: Smoothing splines fit to $n = 50$ data points from $Y = \sin(X) + \epsilon$ on $[0, 2\pi]$ at various λ values. For smaller λ , the fit nearly interpolates all points (wiggly) and has high effective df. As λ grows larger, the fit becomes smoother with lower effective df; for very large λ , it approaches a straight line.

Example code: Smoothing splines in R

```
set.seed(2026)

n <- 50
x <- seq(0, 2*pi, length.out = n)
y <- sin(x) + rnorm(n, sd = 0.2)

df_values <- c(2, 5, 10, 25, 50)
fits <- lapply(df_values, function(df0) {
  smooth.spline(x, y, df = df0)
})

xx <- seq(min(x), max(x), length.out = 400)

preds <- sapply(fits, function(fit) {
  predict(fit, x = xx)$y
})

colors <- c("purple", "blue", "forestgreen", "orange", "red")
```

For additional examples, see [[JWHT21](#), Sec. 7.8.1 & 7.8.2]

```
plot(x, y,
     pch = 16,
     col = "gray40",
     main = "",
     xlab = "X",
     ylab = "Y",
     cex.axis = 1.4,
     cex.lab = 1.4,
     ylim = range(y, preds))

for (i in seq_along(df_values)) {
  lines(xx, preds[, i],
        col = colors[i],
        lwd = 2)
}

legend("topright",
      legend = paste0("df = ", df_values),
      col = colors,
      lwd = 2,
      bty = "n",
      cex = 0.9)
```

Pop-up quiz: Smoothing splines

Suppose a smoothing spline is fit to n observations with distinct x_i 's.

Question: Which statement is most accurate?

- A) The solution has $n + 2$ effective degrees of freedom because it is a natural cubic spline with n observed x_i 's.
- B) The solution is a natural cubic spline with knots at the observed x_i 's, but λ controls the effective flexibility.
- C) Increasing λ adds more knots and makes the fitted curve more wiggly.
- D) When $\lambda \rightarrow \infty$, the smoothing spline becomes a step function.

Answer: B. The smoothing-spline solution is a natural cubic spline with boundary knots at the smallest/largest x_i 's and interior knots at the remaining x_i 's. Its full spline space has nominal df n , and the penalty controls the effective df: from n when $\lambda = 0$ toward 2 as $\lambda \rightarrow \infty$.

Transition: From supervised to unsupervised learning

So far: Supervised learning

$(X, Y) \implies$ learn how X predicts or explains Y .

Now: Unsupervised learning

$X = (X_1, \dots, X_p) \implies$ discover structure in X without a response Y .

Key difference: There is no response variable telling us what the “right answer” is.

New questions:

- Can we summarize high-dimensional data using fewer variables?
- Can we find natural groupings among observations?

Unsupervised learning

Examples:

- Group patients using gene-expression profiles.
- Summarize thousands of measurements with a few new variables.
- Visualize high-dimensional observations in two dimensions.
- Segment customers based on behavior or purchases.

Important caveat: Unsupervised learning is often exploratory; there may not be a single objectively “correct” answer.

Two major tools we will study:

- **PCA for dimension reduction:** find low-dimensional directions capturing most variation.
- **Clustering for grouping observations:** find groups of similar observations.

Principal component analysis (PCA): Overview

Goal: Reduce $X = (X_1, \dots, X_p)$ from p variables to $r \ll p$ new variables while preserving as much variation in X as possible

Reasons for dimensionality reduction:

- *Preprocessing for modeling:* Reduce the number of variables before downstream analysis
- *Computational benefits:* Easier/faster to store and process fewer variables
- *Visualization:* Plotting or interpreting data in 2D or 3D
- *Noise reduction:* Focusing on major signals in the data

How does PCA work?

- Creates new variables, called *principal component scores*
- Each score is a projection of the data onto a direction of high variance
- These directions are called the *principal component directions*, or *principal components* (PCs)

PCA: Visual Illustration for $p = 2$

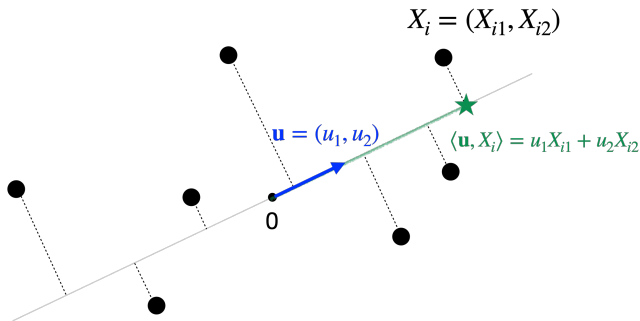


Figure: PCA finds the direction $\mathbf{u}$ such that the projections of the data onto that direction have the largest variance.

PCA as a $2D \rightarrow 1D$ projection

- For any unit vector $\mathbf{u} = (u_1, u_2)$, the projection of an observed point $\mathbf{x}_i$ onto the line spanned by $\mathbf{u}$ is

$$z_i = \langle \mathbf{u}, \mathbf{x}_i \rangle = u_1 x_{i1} + u_2 x_{i2}.$$

- The first PC is the direction (line) where projected points are most spread out; the variance of $z_i = \langle \mathbf{u}, \mathbf{x}_i \rangle$ is maximized
- Geometrically, it is the "major axis" of the data cloud

Toy example: Finding the direction of largest spread (1/2)

Example

Consider a small 2D dataset:

$$\mathcal{X} = \{(-2, -1), (0, 0), (2, 1)\}.$$

These three points lie on the line spanned by $(2, 1)$ and are already centered: their sample mean is $(0, 0)$.

Compare the variances along three directions:

$$\mathbf{u}_x = (1, 0), \quad \mathbf{u}_y = (0, 1), \quad \mathbf{u}_* = \frac{1}{\sqrt{5}}(2, 1).$$

Projection: For a unit vector $\mathbf{u} = (u_1, u_2)$, the projection of $X_i = (x_{i1}, x_{i2})$ onto (the line spanned by) $\mathbf{u}$ is

$$\langle \mathbf{u}, X_i \rangle = u_1 x_{i1} + u_2 x_{i2}.$$

- If $\mathbf{u} = (1, 0)$, the variance in this direction is $\frac{1}{3}((-2)^2 + 0^2 + 2^2) = \frac{8}{3}$.
- If $\mathbf{u} = (0, 1)$, the variance in this direction is $\frac{1}{3}((-1)^2 + 0^2 + 1^2) = \frac{2}{3}$.
- If $\mathbf{u} = \frac{1}{\sqrt{5}}(2, 1)$, the variance in this direction is $\frac{1}{3}((-\sqrt{5})^2 + 0^2 + \sqrt{5}^2) = \frac{10}{3}$.

Toy example: Finding the direction of largest spread (2/2)

Example (Continued)

Projected variances:

Direction	Variance of projected data
$(1, 0)$	$8/3$
$(0, 1)$	$2/3$
$\frac{1}{\sqrt{5}}(2, 1)$	$10/3$

Thus, among the three directions, the variance is maximized along

$$\mathbf{u}_* = \frac{1}{\sqrt{5}}(2, 1).$$

Question: How can we formulate and find the maximum-variance direction?

PCA formulation in $p = 2$

Assumption: The data are centered, i.e. $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i = \mathbf{0}$.

Key idea of PCA:

- For a *direction* $\mathbf{u} = (u_1, u_2)$ with $\|\mathbf{u}\|^2 = u_1^2 + u_2^2 = 1$, the projected value of observation i is

$$z_i(\mathbf{u}) = \langle \mathbf{u}, \mathbf{x}_i \rangle = u_1 x_{i1} + u_2 x_{i2}.$$

- The empirical variance¹ of the projected scores along $\mathbf{u}$ is

$$\text{Var}(\langle \mathbf{u}, \mathbf{X} \rangle) = \frac{1}{n} \sum_{i=1}^n (z_i(\mathbf{u}) - \bar{z}(\mathbf{u}))^2 = \frac{1}{n} \sum_{i=1}^n \langle \mathbf{u}, \mathbf{x}_i \rangle^2 = \frac{1}{n} \sum_{i=1}^n (u_1 x_{i1} + u_2 x_{i2})^2.$$

- The **1st principal component** is the direction $\mathbf{u}$ that *maximizes* this variance:

$$\text{maximize } \frac{1}{n} \sum_{i=1}^n \langle \mathbf{u}, \mathbf{x}_i \rangle^2 \quad \text{subject to} \quad \|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2} = 1.$$

- In words:** choose the unit direction whose projected scores have the largest variance.

¹We use the PCA convention of computing variance with denominator n .

General formulation of PCA beyond 2 dimensions

First PC: a unit vector $\mathbf{u}_1 \in \mathbb{R}^p$ that maximizes variance, i.e.,

$$\mathbf{u}_1 = \operatorname{argmax}_{\|\mathbf{u}\|=1} \frac{1}{n} \sum_{i=1}^n (\mathbf{u} \cdot \mathbf{x}_i)^2$$

Second PC: a unit vector $\mathbf{u}_2 \in \mathbb{R}^p$ maximizing variance, **subject to being orthogonal to $\mathbf{u}_1$** ,

$$\mathbf{u}_2 = \operatorname{argmax}_{\substack{\|\mathbf{v}\|=1 \\ \langle \mathbf{v}, \mathbf{u}_1 \rangle = 0}} \frac{1}{n} \sum_{i=1}^n (\mathbf{v} \cdot \mathbf{x}_i)^2$$

- $\mathbf{u}_1 \perp \mathbf{u}_2 \implies$ the random variables $Z_1 = \langle \mathbf{u}_1, \mathbf{X} \rangle$ and $Z_2 = \langle \mathbf{u}_2, \mathbf{X} \rangle$ are *uncorrelated*

Subsequent PCs $\mathbf{u}_3, \dots, \mathbf{u}_p$ are defined analogously, each orthogonal to all preceding PCs

Interpretation:

- The k -th PC is orthogonal to all prior ones, ensuring uncorrelatedness among the PC scores
- (Optional) The PC directions correspond to the eigenvectors of the sample covariance matrix

PCA illustration 2: $p = 3$ to $r = 2$

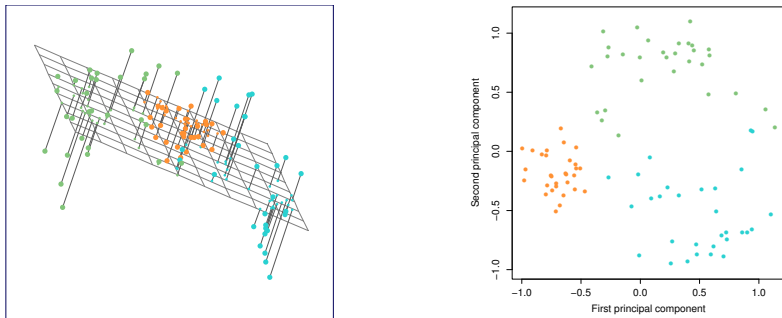


Figure: Ninety observations in $\mathbb{R}^3$. **Left:** The first two PC directions span a plane that best fits the data, minimizing total squared distance. **Right:** Data are “flattened” onto that 2D plane, forming their PC scores [JWHT21, Figure 12.2].

Key idea in higher dimension:

- Find a subspace of dimension r that captures maximal variance (=minimizing residuals)
- $\mathbf{u}_1$ is the top PC direction, $\mathbf{u}_2$ is second, etc., each orthogonal

PCA and standardization

PCA is based on variance. Variables with larger numerical scale can dominate the first principal components.

Example: Without standardization, X_1 may dominate simply because it has larger units.

- X_1 = income measured in dollars.
- X_2 = age measured in years.

Common practice of standardization:

$$\tilde{X}_j = \frac{X_j - \bar{X}_j}{s_j}.$$

- Centering is essential: PCA studies variation around the mean.
- Standardization is often used when variables are measured on different scales or have very different variances.

Pop-up quiz: PCA intuition

Suppose a centered two-dimensional dataset has the following projected variances along three unit directions:

Direction	$(1, 0)$	$(0, 1)$	$\frac{1}{\sqrt{2}}(1, 1)$
Projected variance	2	1	5

Question: Which statement is most accurate?

- A) The first PC among these directions is $(0, 1)$, because it has the smallest variance.
- B) The first PC among these directions is $\frac{1}{\sqrt{2}}(1, 1)$, because it has the largest projected variance.
- C) PCA chooses the direction that best predicts a response variable Y .
- D) PCA does not require centering or scaling under any circumstances.

Answer: B. PCA chooses directions of large variance in X , not directions that predict Y . Centering is essential, and scaling may be important when variables have different units.

Wrap-up

- **Smoothing splines:** The solution is a natural cubic spline with knots at the observed x_i 's, but λ controls the effective flexibility.
 - **Effective degrees of freedom:** For distinct x_i 's, $\lambda = 0$ gives effective df n , while $\lambda \rightarrow \infty$ gives effective df 2.
- **Unsupervised learning:** We study structure in X without a response Y .
- **PCA:** Finds directions of maximum variance and uses projections onto those directions for dimension reduction.
 - **PCA scores:** Each observation is represented by its projections onto PC directions; keeping the first few scores gives a lower-dimensional representation.
 - **Standardization matters:** Since PCA is based on variance, variables with larger scales can dominate unless variables are standardized.

Suggested reading: [JWHT21, Ch. 12.1–12.2]

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.